

CO5_AT1 CONSTRAINT-BASED PROBLEM SOLVING

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1. Given a set of items with different sizes and a fixed bin capacity, implement a solution to pack all items into the minimum possible number of bins without exceeding the capacity of any bin. Use a backtracking-based or heuristic approach with pruning to eliminate infeasible packings. Analyze the constraints, packing efficiency, and NP-Hard complexity of the problem.

CODE:

```
3  def bin_packing(items, capacity):
9      for item in items:
13         for i in range(len(bins)):
14             if sum(bins[i]) + item <= capacity:
15                 bins[i].append(item)
16                 placed = True
17                 break
18
19         # Create a new bin if item cannot fit
20         if not placed:
21             bins.append([item])
22
23     return bins
24
25
26 # Input
27 items = [4, 8, 1, 4, 2, 1]
28 capacity = 10
29
30 # Check feasibility
31 if any(item > capacity for item in items):
32     print("Packing not possible: An item exceeds bin capacity.")
```

OUTPUT:

Output

```
Minimum bins used (heuristic): 2  
Bin 1: [8, 2], Used = 10/10  
Bin 2: [4, 4, 1, 1], Used = 10/10  
  
=== Code Execution Successful ===
```